

GUIDANCE COUNSELING MANAGEMENT SYSTEM FOR CATSU

A Capstone Project
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CHAPTER 1

INTRODUCTION

Project Context

Guidance counseling is a vital process that helps students understand themselves, make informed decisions, and develop the skills necessary to handle personal, academic, and social challenges. It supports students' holistic development by guiding them in achieving their educational and personal goals. According to Alegado and Alcantara (2021), guidance counseling contributes significantly to students' success by enhancing their self-awareness, personality, and emotional well-being.

At Catanduanes State University (CatSU), the Guidance and Counseling Office (GCTO) continue to provide services that help students cope with academic pressure, career uncertainties, and personal concerns. However, the current **walk-in appointment system** often leads to scheduling conflicts, manual record-keeping errors, and longer processing times. These inefficiencies can delay the delivery of timely support and increase administrative workload for guidance staff.

To address these challenges, this study proposes the development of a **Guidance Counseling Management System (GCMS)** — a web-based platform designed to **automate scheduling, digitalize case documentation, and generate counseling reports**. The system aims to minimize paperwork, reduce scheduling conflicts, and enhance

accessibility to counseling services. As highlighted by Abiog et al. (2024), adopting digital systems in guidance offices can significantly improve efficiency, data management, and service delivery.

Through the GCMS, students will be able to easily book appointments online, while counselors can manage records and monitor student progress more effectively. This initiative supports CatSU's goal of providing student-centered, technology-driven services that promote academic success and overall well-being.

Purpose and Description

The **Guidance and Counseling Office (GCTO)** of Catanduanes State University (CatSU) currently relies on a manual, walk-in process for managing student counseling services. This traditional system often leads to scheduling conflicts, lost records, and delays in assisting students who require immediate guidance. The manual handling of appointment forms and counseling data also increases the workload of guidance staff, making it difficult to efficiently track student progress and maintain organized records.

To address these limitations, this study aims to develop a **web-based Guidance Counseling Management System (GCMS)** that automates the scheduling and record-keeping processes. The system integrates core features such as online appointment booking, digital case management, and automated report generation. By transitioning to a

digital platform, the GCTO can streamline its workflow, ensure data accuracy, and provide a more organized and efficient service delivery for both students and counselors.

Furthermore, the system seeks to enhance accessibility and transparency in guidance operations. Students will be able to schedule counseling sessions conveniently online, while counselors can monitor and update case files in real time. The system's data-driven reporting capability will also support administrators in evaluating counseling trends and improving service outcomes. Overall, the GCMS aligns with CatSU's commitment to delivering innovative, student-centered services that promote academic success and personal development.

Objectives

The primary objective of this project is to design and implement an automated **Guidance Counseling Management System (GCMS)** that simplifies the process of booking and managing counseling sessions.

Specifically, this study aims to:

1. Develop an intuitive and efficient web-based scheduling system that automates appointment booking for students and counselors.
2. Implement digital case management for accurate record-keeping and monitoring of counseling sessions.

3. Generate reports summarizing counseling activities to support decision-making and service evaluation.
4. Evaluate the system's quality and effectiveness using the **ISO/IEC 25010 software quality model**, focusing on functionality, usability, performance efficiency, reliability, and security.

Scope and Limitations

The scope of this study covers the design, development, and implementation of a web-based Guidance Counseling Management System for CatSU. The system's core modules include appointment scheduling, counselor case management, and report generation. It caters to three user roles: students, counselors, and administrators. The GCMS provides online booking features, automated notifications, and centralized data storage to enhance accessibility, efficiency, and service monitoring.

The study is limited to the system's technical and operational aspects within CatSU. It does not measure the psychological effectiveness of counseling sessions but focuses on the functionality and usability of the platform. The system's deployment is site-specific, meaning findings may not fully apply to other institutions. Furthermore, its performance depends on the university's IT infrastructure and the users' adaptability to digital systems.

Definition of Terms

This section defines key terms used in the study to ensure clarity and consistency in interpreting the concepts and processes involved in the Guidance Counseling Management System.

Case Management – A system feature that allows counselors to document, organize, and monitor students' counseling histories and progress.

Compatibility – The ability of the system to function effectively with other software, hardware, or environments.

Functional Suitability – The extent to which the system provides functions that meet the specified user needs.

Usability – The ease with which users can learn, navigate, and interact with the system.

Performance Efficiency – The system's ability to perform tasks quickly and effectively while utilizing resources optimally.

Reliability – The consistency of the system's performance over time without failures.

Security – The capability of the system to protect information and prevent unauthorized access.

Maintainability – The ease of modifying or updating the system to correct issues or enhance performance.

Portability – The ability of the system to be deployed or adapted across different environments or platforms.

ISO/IEC 25010 – An international standard for evaluating software quality through eight key characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Conceptual Framework

The conceptual framework serves as the foundation for understanding how the **Guidance Counseling Management System (GCMS)** operates and achieves its objectives. It illustrates the logical relationship between the key components of the study from the resources needed for development to the processes involved and the expected outcomes. This framework provides a structured overview of how the system is designed, developed, and evaluated to improve the efficiency of guidance counseling services at Catanduanes State University (CatSU).

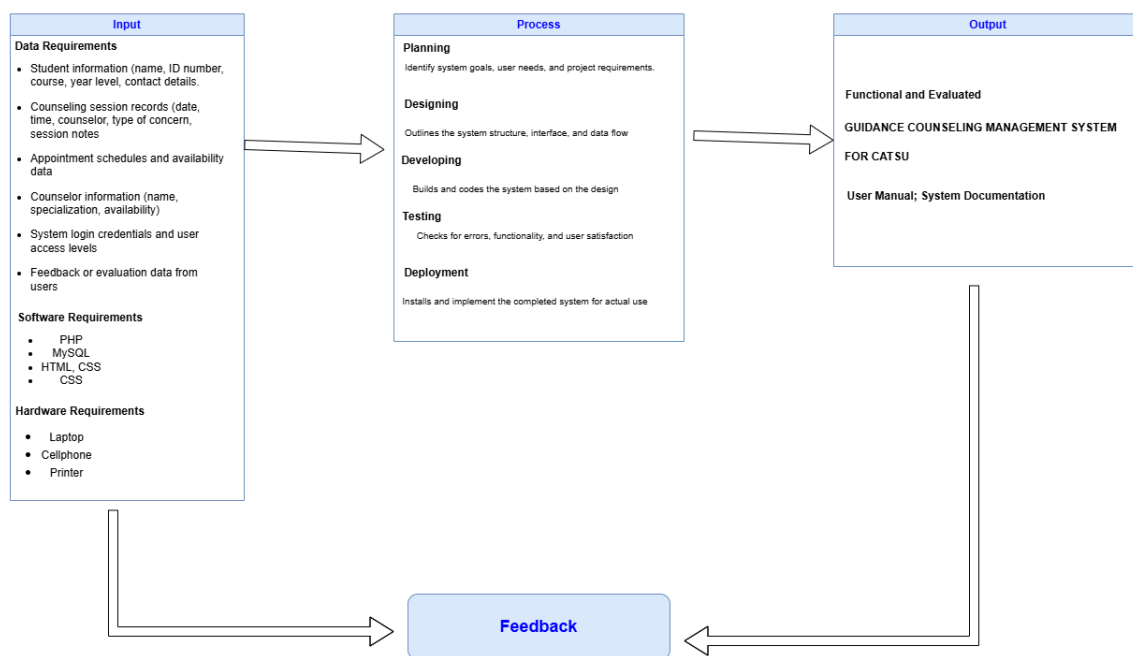


Figure 1. Conceptual Framework

This study is structured using the **Input-Process-Output (IPO) model**, which shows the logical flow of activities and resources throughout the development of the system. The **Input** consists of the necessary requirements such as student and counselor information, appointment schedules, and other related data, along with the software and hardware tools needed for system creation and operation. The **Process** includes the key stages of system development—**planning, designing, developing, testing, and deployment**—where each phase plays an important role in transforming inputs into a functional output. Planning identifies user needs and objectives; designing structures the system's interface and database; developing involves the actual coding and system

construction; testing ensures accuracy and usability; and deployment implements the system for actual use.

The **Output** of the framework is the fully developed **Guidance Counseling Management System**, which automates counseling appointment scheduling, digital record-keeping, and report generation. The system aims to reduce paperwork, minimize scheduling conflicts, and enhance the accessibility of counseling services to students. By establishing a systematic flow of processes through the IPO model, the framework ensures that every aspect of system development aligns with the main goal of the study — to improve the efficiency, organization, and responsiveness of guidance and counseling operations at CatSU.

CHAPTER 2

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter presents a review of related literature, studies, and technologies relevant to the development of the **Guidance Counseling Management System (GCMS)**. It discusses various concepts, models, and previous research that provide the foundation for understanding how guidance and counseling services can be improved through digital innovation. The review covers studies on the management and organization of guidance services, the integration of information and communication technologies in counseling systems, and the evaluation of web-based platforms that enhance accessibility and efficiency. By examining both local and international research, this chapter identifies the trends, similarities, and gaps in existing systems, which serve as the basis for developing a localized GCMS tailored to the needs of Catanduanes State University.

Guidance Counseling Management System

Kartadinata (2022) conducted a study titled *Quality Improvement and Management System Development of School Guidance and Counseling Services*, published in the **Journal of Universitas Negeri Malang**. The study developed and evaluated a model to improve the quality of guidance and counseling services across various school levels. By introducing a computerized management framework, the research highlighted the

significance of systematizing guidance services through structured procedures and digital recordkeeping. This aligns with the present study's objective to develop a centralized digital system that supports efficient management of counseling sessions and student information.

Fataroni (2023), in *Implementation of Guidance and Counseling Service Management in an Effort to Improve the Quality of Student Learning* from **Bale Literasi**, examined how the management functions of planning, organizing, leading, and controlling contribute to improving student learning outcomes. The study emphasized that proper management of guidance services enhances not only student performance but also counselor efficiency. This is closely related to the current study, which also focuses on improving management efficiency by automating routine tasks in the counseling office.

Similarly, Karunia et al. (2023) in *Management Strategies for Guidance and Counseling Services in Schools to Support Student Achievement*, published in **Manajia**, presented a literature-based analysis of effective management strategies for school guidance programs. Their work underscores the importance of well-defined processes, counselor coordination, and record organization. These insights support the rationale for developing a structured information system like the GCMS to ensure organized service delivery.

Furthermore, Jannah, Neviyarni, and Sukur (2022) in *The Role of Counselors in Developing Guidance and Counseling Management in Schools* from **Quality Journal**, explored how counselors contribute to designing and maintaining guidance management

systems. The authors found that counselor involvement in system development improves implementation and user acceptance. This supports the inclusion of counselors and staff feedback in the design process of the proposed GCMS, ensuring that the system meets actual user needs.

Technologies Integrated in GCMS

Nasution et al. (2024), in *Integration of Information Technology in Guidance and Counseling Management in Schools: Challenges and Implementation Strategies* published in **Attanwir Journal**, discussed the integration of IT into guidance counseling management. The study identified challenges such as limited digital literacy, technical constraints, and resource availability, but emphasized that proper training and infrastructure lead to more efficient service delivery. This finding supports the technological framework of the proposed GCMS, which integrates IT tools to simplify administrative tasks.

Majjate et al. (2024) proposed an *AI-Powered Academic Guidance and Counseling System Based on Student Profile and Interests*, published in **MDPI**. Their system utilized artificial intelligence to recommend personalized academic guidance. Although more advanced than the proposed GCMS, it demonstrates the growing trend of using intelligent

technologies to improve student support services. This reinforces the idea that automation and data-driven systems enhance the responsiveness and accuracy of guidance operations.

Azmin (2023), in *Integrating ICT into Career Counseling: An Empirical Study of the TACC Digital Content Model in Malaysian Secondary Education* from **HRMars**, investigated the integration of ICT in delivering career counseling services. The study found that digital models improve student engagement and counselor productivity. This supports the use of web technologies in your GCMS, which also aims to increase accessibility and user interaction.

In addition, Setyawan et al. (2023) in *Design of Continuous Web App: Guidance and Counseling Management Information System at SMKN 1 Purwakarta using Laravel Framework*, published in **Lembaga KITA**, developed a dynamic GCMS using the Laravel framework. The study's design approach and use of modern web tools closely parallel the methodology and technologies of your own system.

Likewise, Balmores et al. (2024) created an *Online Individual Inventory and Counseling Information System* for **Cagayan State University–Lasam**, which digitized student records and counseling forms. Their work demonstrated that a localized web-based system effectively streamlines information management and reduces manual documentation — an outcome that directly parallels the goals of your proposed GCMS.

Evaluation of Web-Based GCMS

Sudrajat et al. (2024), in *Evaluating Technology Acceptance in Counseling Services: An Analysis of Principal Indicators Utilizing the UTAUT2 Model via SEM-PLS*, published in EJournal UPI, examined factors affecting technology acceptance among counselors and administrators. The study concluded that performance expectancy and ease of use strongly influence user adoption of digital counseling platforms. This evaluation model can guide future assessments of your GCMS's usability and acceptance.

Guntara et al. (2023), in *Web-Based Counseling Skills Evaluation Information System Using Design Science Research Methodology (DSRM) Approach*, published in IJADIS, developed and assessed a system to evaluate counselor performance. Their use of the DSRM framework demonstrates how system development and evaluation can occur in iterative stages — similar to your project's methodological structure.

Lastly, Bato and Marcial (2016), in *Students' Attitudes Towards the Development of an Online Guidance Counseling System* from IITTL Journal, analyzed student perceptions toward web-based counseling platforms. Findings revealed positive attitudes toward digital accessibility and convenience. This supports the user-centered design of your GCMS, which aims to enhance student engagement through an online interface.

Synthesis

The reviewed studies collectively emphasize the growing importance of technology in improving the management and delivery of guidance and counseling services in educational institutions. Under the first thematic area, researchers such as Kartadinata (2022), Fataroni (2023), and Karunia et al. (2023) highlighted that effective management systems enhance counselor performance, improve service organization, and positively influence student learning. Their findings suggest that structured guidance management, when supported by clear planning and documentation, leads to more efficient and responsive services. Likewise, Jannah et al. (2022) underscored the crucial role of counselors in developing and maintaining such systems, pointing out that user involvement ensures better implementation and sustainability. Together, these studies show that the modernization of counseling services depends not only on management strategies but also on the active participation of those who use the system.

Studies under the second thematic area demonstrate that integrating technology into counseling operations brings substantial benefits but also introduces new challenges. Nasution et al. (2024) and Azmin (2023) both noted that information and communication technologies improve accessibility, accuracy, and efficiency in service delivery, while also requiring adequate training and infrastructure. Majjate et al. (2024) showed the potential of artificial intelligence to personalize counseling, while Setyawan et al. (2023) and Balmores et al. (2024) developed web-based systems using frameworks such as Laravel and PHP/MySQL to digitalize appointment scheduling and record management. These

studies illustrate the steady shift toward digital transformation in guidance services, proving that technological integration enhances both counselor workflow and student engagement.

Meanwhile, the third thematic area centers on the evaluation and user acceptance of web-based guidance systems. Sudrajat et al. (2024) applied the UTAUT2 model to examine user acceptance, showing that perceived usefulness and ease of use significantly influence adoption. Guntara et al. (2023) employed a design science research approach to evaluate system effectiveness and counselor performance, while Bato and Marcial (2016) explored positive student attitudes toward online counseling platforms. These studies confirm that web-based systems are generally well-received and effective when designed around user needs and tested through structured evaluation methods.

Despite these advancements, a noticeable **research gap** remains. Most existing systems focus on either specific counseling functions or the evaluation of technology use but rarely combine both development and implementation in a localized academic context. Few studies address the unique challenges of integrating an all-in-one digital platform for managing counseling schedules, student cases, and reports within a single institution. The present study addresses this gap by developing a comprehensive, web-based **Guidance Counseling Management System (GCMS)** tailored for Catanduanes State University. Unlike previous works, it unifies administrative management, appointment scheduling, and documentation processes into one accessible system—providing an innovative, student-centered solution that modernizes the university's guidance operations

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technical background of the study, which includes the hardware, software, data, and user requirements needed for the development and operation of the **Guidance Counseling Management System (GCMS)** for Catanduanes State University. These components serve as the technical foundation of the system, ensuring that it functions effectively and meets the needs of its intended users.

Hardware Requirements

The hardware components provide the physical infrastructure necessary to develop, test, and deploy the GCMS. Adequate hardware ensures smooth operation, reliable performance, and accessibility for both developers and users in the Guidance and Counseling Office.

Table 1. Hardware Requirements

Hardware Requirement	Specification	Description
Laptop	Intel Core i5-1335U (13th Gen), 8GB RAM, 512GB SSD, 15.6" FHD, Wi-Fi 6E, Windows 11, Cypress Green	Used by developers and staff for system development, testing, and operation. Ensures smooth performance and multitasking..

Printer	Epson L3110, Print/Scan/Copy functions	Used to print reports, forms, and other official documents from the system.
Router	PLDT Home Fiber, Wireless LAN	Provides network connectivity for accessing the GCMS via the internet or intranet.

The hardware requirements listed above provide the necessary resources for system development and implementation. The laptop or desktop computer serves as the main platform for programming, debugging, and testing. The printer allows counselors to generate and print reports and other documentation, while the router ensures stable internet connectivity, enabling users to access the web-based GCMS anytime and anywhere within the school network.

Software Requirements

The software requirements consist of the programs, tools, and environments used for coding, designing, testing, and deploying the system. These software components ensure that the GCMS operates efficiently and can be accessed via any compatible web browser.

Table 2. Software Requirements

Software Requirement	Specification	Description
Operating System	Windows 11	Manages computer operations and coordinates processing of system applications.
Web Server	Apache	Hosts and runs the GCMS locally for development and testing.
Database Management System	MySQL	Stores and manages data including student profiles, counselor information, and appointment records.
Programming Language	PHP, HTML, CSS, javascript	Used for developing system functionalities, web design, and interactive features.
Web Browser	Google Chrome	Used to access and test the GCMS interface and functionality.
Office Tools	Microsoft Office	Used for documentation and preparation of reports.
Localhost Environment	Xampp	Provides an integrated platform for local web server deployment and testing.

The software requirements are essential for developing a stable and interactive web-based platform. XAMPP provides a local server environment where the system can be developed and tested using Apache and MySQL. The programming languages PHP, HTML, CSS, and JavaScript ensure that the GCMS is functional, visually appealing, and user-friendly. Web browsers such as Chrome and Firefox allow the team to test the

responsiveness and compatibility of the system, while office tools are used for documentation and reporting.

Data Requirements

The data requirements identify the specific information that the GCMS will store, process, and manage. These data sets are crucial for providing accurate records, generating reports, and supporting decision-making within the Guidance and Counseling Office.

Table 3. Data Requirements

Construct/ Variables	Source of Data	Data Gathering Technique	Instrument	Data Processing
Counselor Officials Information	Guidance Office	Document Analysis	Documentary Review	Input and validation in the database
Student Profile	Student	Interviews and Surveys	Questionnaire	Structured data input in the system
Appointment Schedule	Guidance Office	Observation / Interview	Interview Guides	Data encoding and automated tracking
Counseling Records	Guidance Office	Interviews	Structured Form	Digital documentation and reporting
System Evaluation	Users (students, counselors, admin)	Descriptive Survey Method	Questionnaire	Mean and frequency count for analysis

The table outlines the essential data collected and processed by the system. Counselor and student information form the core of the GCMS database, allowing efficient tracking of sessions and appointments. Appointment schedules and counseling records ensure that each student interaction is recorded systematically. The evaluation data gathered through surveys and interviews provide valuable feedback that helps improve the system's usability and functionality.

User Requirements

The user requirements define the system functionalities and access privileges for each type of user administrators, counselors, and students. These requirements serve as the blueprint for the GCMS design and ensure that each user role performs its intended tasks efficiently.

Table 4. User Requirements

Module	Function	Description	User Role
Dashboard	Overview of System Status	Displays key metrics such as number of active users, appointment reports	Admin
Appointment Management	Schedule and Track sessions	Allows admin to set manage and track appointments	Admin
Student Management	View and Update Student Records	Add update or remove student records	Admin
Counseling Management	Manage student counseling Information	Organizing student counseling information	Admin, Counselor

SMS Notification	Send and Receive Notification	Send and receive for counselors from client	Admin, Counselor
Generate Report	Counseling	Produce reports based on session	Admin, Counselor

The table shows the user requirements needed for the system, a digital platform intended to streamline how students access and receive support. For students, its easy ways to book appointments and manage their counselling journey, for counsellor and staff, it provides a central, secure place to organized student profile, keep track of session notes, and generate important reports, replacing the hassle of manual records-keeping. The system also includes a dashboard for administrator, offering a quick and clear overview of counselling activities and the overall effectiveness of the service.

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter discusses the design, development, and methodology applied in creating the **Guidance Counseling Management System (GCMS)** for Catanduanes State University. It details the requirements analysis, documentation, design of software and systems, development, testing, respondents, and statistical tools used for system evaluation.

The software development methodology applied in this project is the **Agile Methodology**, which emphasizes iterative development, collaboration with stakeholders, and continuous improvement. Agile was selected because it allows flexibility in addressing user feedback and ensures that the system is aligned with the needs of the Guidance Counseling and Testing Office (GCTO).

Requirements Analysis

The requirements were gathered through multiple techniques, including interviews with guidance counselors, surveys with students, and direct observations of existing counseling processes at CatSU. The current process is largely paper-based, requiring

students to manually fill out consent forms and book counseling appointments through walk-ins. This creates inefficiencies such as scheduling conflicts, missed appointments, and difficulties in record management.

The proposed system addresses these issues by introducing a digital platform for booking appointments, managing counseling records, and generating reports. The data collected through requirement analysis provided the foundation for both functional and non-functional requirements.

Requirements Documentation

The requirements for the Guidance Counseling Management System are presented in tabular form below, categorized into functional, non-functional, assumptions, and constraints, each followed by a discussion.

Table 5. Functional Requirements

REQUIREMENTS	DESCRIPTIONS
Authentication & Security	Users (Student, Counselor, Admin) must log in securely. Passwords must be hashed

	using Bcrypt encryption, and sessions must be role-validated.
Student Appointment & Wellness	Students must be able to book appointments and are required to answer a Wellness Questionnaire (Physical, Intellectual, Environmental) during the booking process.
Counselor Session Management	Counselors must be able to view appointments, read student wellness responses, and record confidential Session Notes and recommendations in the system.
Administrative Management	Administrators must manage user accounts, oversee the master schedule, and assign counselors to pending appointments if necessary.
Notification System	The system must send automated in-system and email alerts (via SMTP) to students and counselors regarding appointment status updates.

Table 5 functional requirements ensure that the core features directly support the daily activities of students, counselors, and administrators. A key improvement in this system is the integration of the **Wellness Questionnaire**, which requires students to assess their Physical, Intellectual, and Environmental wellness during the booking process. This feature provides counselors with immediate, actionable context before the session begins, a capability that was absent in the manual process.

Furthermore, the automated Notification System addresses the issue of missed appointments by triggering real-time email alerts via PHPMailer whenever a schedule is requested or confirmed. The strict Authentication & Security module ensures that sensitive student records are compartmentalized, allowing access only to authorized counselors and administrators, thereby streamlining workflow while strictly maintaining confidentiality.

Table 6. Non-Functional Requirements

REQUIREMENTS	DESCRIPTIONS
Usability	The system interface must be responsive and user-friendly, utilizing the Bootstrap 5 framework to ensure compatibility across desktops and tablets.

Security	Data privacy must be enforced through PDO Prepared Statements to prevent SQL injection, and session hijacking protection must be active..
Reliability	The system must maintain data integrity for student records and ensure email notifications are delivered via the integrated mail server.
Performance	The system should process appointment requests and load dashboard analytics within three seconds.

Table 6 Non-functional requirements define the quality attributes that ensure the system is effective and sustainable. Usability is significantly enhanced through the use of the Bootstrap 5 framework, which provides a responsive, grid-based layout that adapts seamlessly to both desktop and tablet screens, ensuring accessibility for all students.

Security is a critical priority; the system safeguards data by using Bcrypt hashing for passwords and PDO Prepared Statements for database interactions. This technical approach actively prevents common threats such as SQL injection and unauthorized account access. Finally, Performance standards ensure that the system remains responsive

(loading in under 3 seconds) even during peak usage, such as examination periods when counseling requests may spike.

Design of Software, Systems, Product and/or Processes

Context Diagram

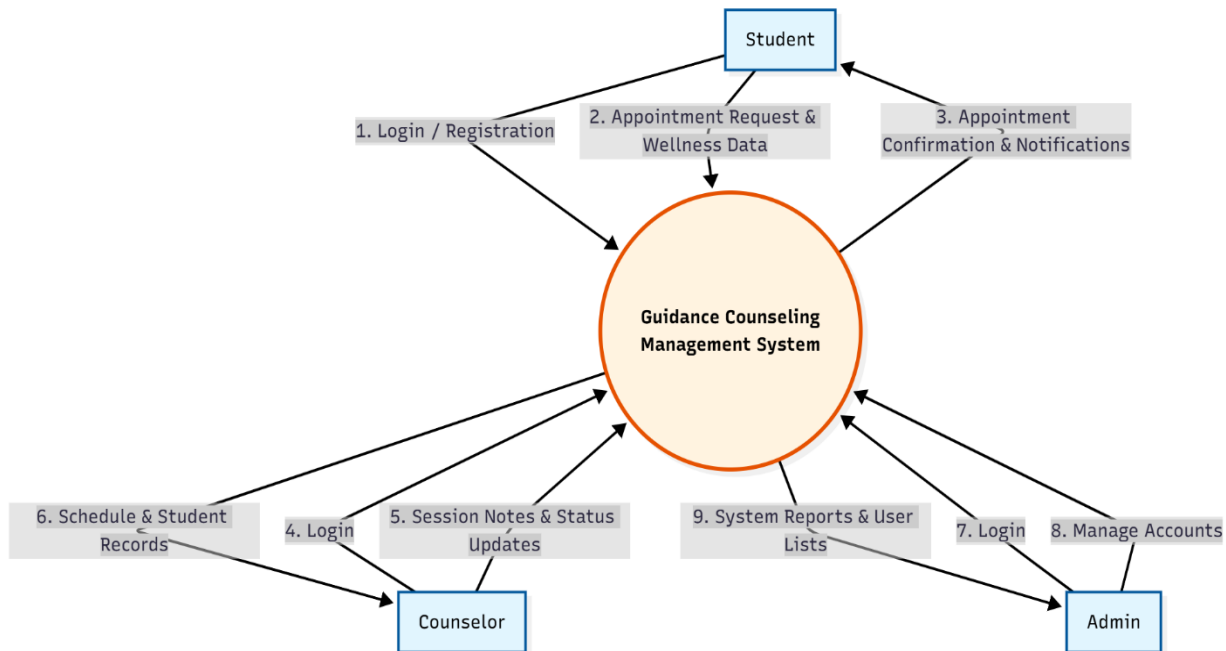


Figure 2. Context Diagram

Figure 2 illustrates the Guidance Counseling Management System (GCMS) as the central processing unit that mediates all interactions between the system and its three primary external entities: the Student, the Counselor, and the Admin. As shown in the

diagram, the Student acts as the primary service seeker who accesses the system to perform secure login or registration, submit appointment requests accompanied by wellness data, and receive real-time notifications regarding the status of their bookings. The Counselor, serving as the service provider, interacts with the system to input session notes and update appointment statuses, while the system reciprocates by providing updated schedules and access to essential student records. Overseeing these operations is the Admin, who manages user accounts and monitors the overall system performance through generated reports and user lists. Together, these data flows demonstrate how the GCMS functions as a centralized hub that streamlines the scheduling, documentation, and management of guidance services.

DATA FLOW DIAGRAM

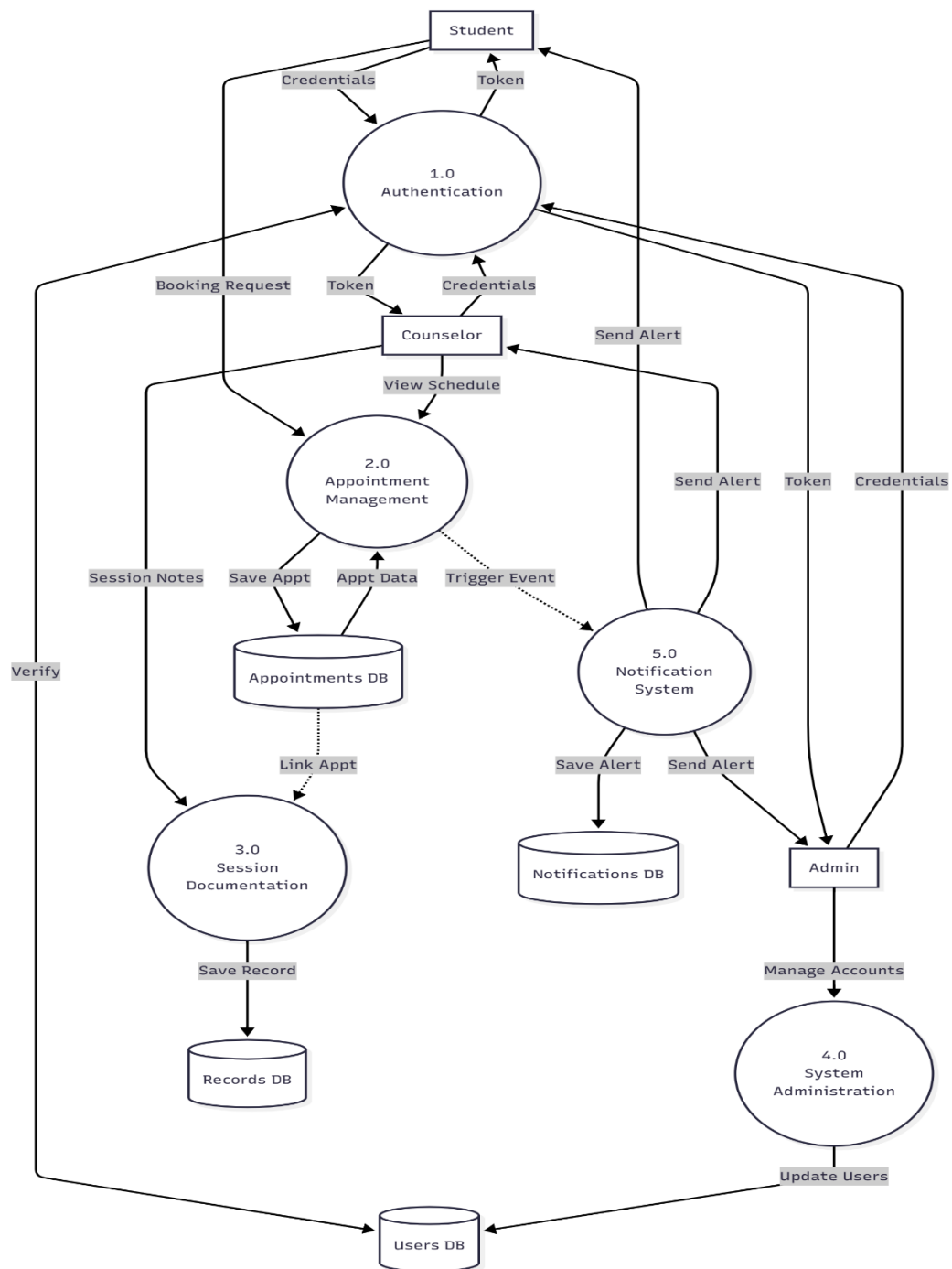


Figure 3. Data Flow Diagram Level 1 (Student, Counselor, Admin)

Figure 3 depicts the Level 1 Data Flow Diagram (DFD) of the Guidance Counseling Management System, illustrating the logical flow of information between the system's external entities and its core processing modules. The diagram organizes the system into five primary processes: 1.0 Authentication, 2.0 Appointment Management, 3.0 Session Documentation, 4.0 System Administration, and 5.0 Notification System.

The data flow begins at 1.0 Authentication, where the Student, Counselor, and Admin entities submit their login credentials to be verified against the Users Database. Once authenticated, Students interact with 2.0 Appointment Management to submit booking requests. These requests are stored in the Appointments Database and simultaneously trigger 5.0 Notification System. This notification process generates real-time alerts (via email or in-system updates), saves them to the Notifications Database, and broadcasts them to the Student (confirmation), Counselor (new request), and Admin (monitoring alert).

Counselors also access the appointment module to manage their schedules before moving to 3.0 Session Documentation, where confidential session notes are securely stored in the Counseling Records Database. Parallel to these operations, the Admin entity interacts with 4.0 System Administration to manage user accounts and system configurations.

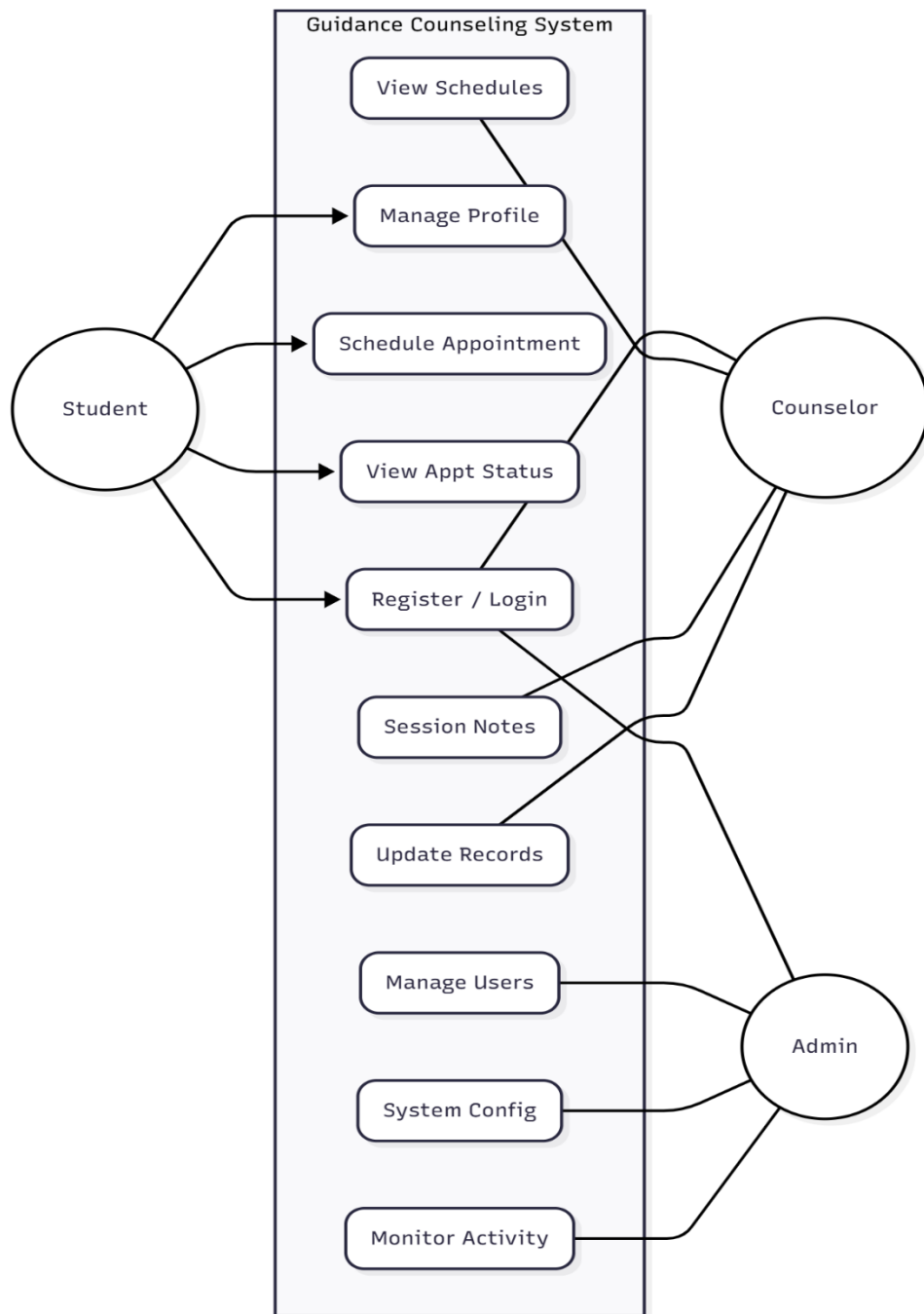
USE CASE DIAGRAM

Figure 4. Use Case Diagram

Figure 4 presents the Use Case Diagram for the Guidance Counseling Management System (GCMS), illustrating the interaction between the system and its three primary actors: the Student, the Counselor, and the Administrator. The diagram defines the boundaries of the system and details the specific functions each actor performs to ensure a streamlined and digitalized counseling process.

The Student is the primary end-user of the system. Their interaction begins with the Registration and Login process, where they secure access to the platform using their credentials. Once authenticated, students engage with the Profile Management module, which allows them to update their personal information. Crucially, the system requires students to complete a Wellness Intake Form—covering Physical, Intellectual, and Environmental wellness—before booking an appointment.

The core functionality for the student is Schedule Appointment. Through this use case, students view the availability of counselors, select a preferred date and time, and submit a booking request. The system initially flags these requests as "Pending." Students can then use the View Appointment Status function on their dashboard to track whether their request has been confirmed, rescheduled, or declined by the counselor, ensuring they are always informed about their counseling schedule without the need for physical follow-ups.

The Counselor acts as the service provider within the system. After logging in, their primary workflow centers on the Manage Schedule use case. Unlike manual systems, the

GCMS provides counselors with a dedicated "My Schedule" dashboard where they can view all incoming appointment requests. They have the authority to Confirm or Cancel appointments directly, which automatically updates the student's status notification.

A critical function for this actor is Session Management. During or after a counseling session, the counselor uses the system to retrieve the student's wellness profile and appointment history. They are responsible for performing the Record Session Notes task, where they document the session's outcome, key discussions, and recommendations. These notes are securely stored in the database, allowing counselors to track a student's progress over time. Additionally, counselors can Update Student Records, ensuring that any changes in a student's status or needs are accurately reflected in the system.

The Administrator serves as the system overseer, ensuring operational stability and security. Their interaction focuses on User Account Management, where they create and maintain accounts for counselors and verify student registrations. This ensures that only authorized personnel and bona fide students access the system.

The Administrator also utilizes the Monitor System Activity use case via the Admin Dashboard. This module provides a high-level view of the guidance office's performance, displaying critical metrics such as the total number of appointments, the count of active counselors, and pending referrals. Furthermore, the admin has the capability to Configure System Settings and oversee global appointment flows, allowing them to intervene in

scheduling conflicts or reassign students to different counselors if necessary to maintain an efficient service delivery.

Activity Diagram

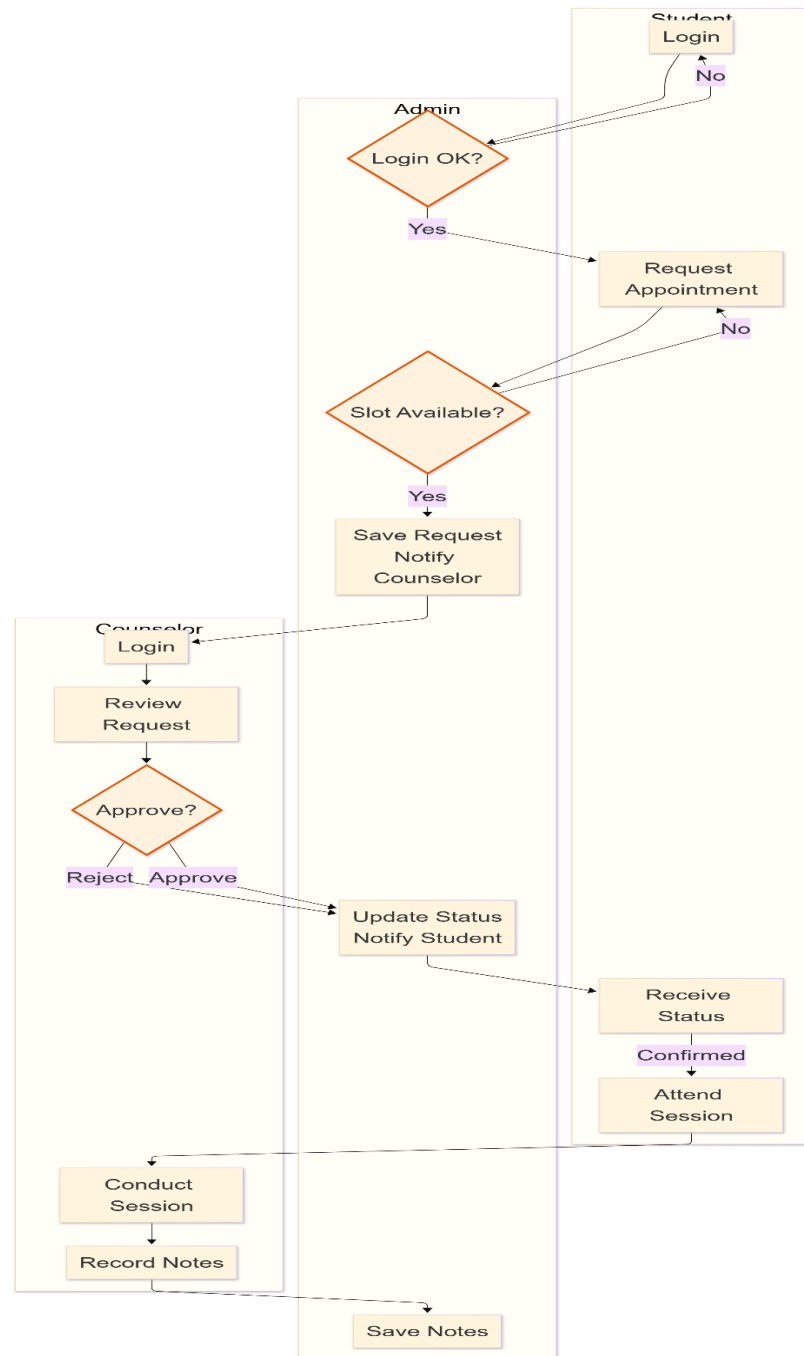


Figure 5. Activity Diagram/Process Swimlane.

The figure shows the swimlane diagram of the Guidance Counseling Management System, which illustrates the responsibilities and flow of interactions among the three main actors: Student, Counselor, and Admin, together with the System. Each lane defines the role of a particular actor, while the arrows demonstrate the sequence of processes until all actions are finalized and stored by the system in the database.

For the Student, the process begins with login or registration. The System authenticates the user's credentials, and if authentication fails, an error message is displayed, thereby terminating the process. If authentication is successful, the student proceeds to schedule or reschedule an appointment. The System processes this request and updates the counselor's appointment list. The counselor can then view the scheduled appointments through the system, after which the System stores and updates the appointment information in the database.

For the Counselor, the process also begins with login. The System authenticates the counselor's credentials, and upon successful login, the counselor can access and view the appointments. The counselor may then record session notes and update student records after counseling sessions. These updates are received by the System, which stores the information into the appropriate databases, such as counseling records and student information, ensuring that data is accurate and securely maintained.

For the Admin, the process starts with login as well. The System verifies the admin's credentials, and upon successful authentication, the admin can perform administrative functions, including managing user accounts and configuring system settings. All changes made by the admin are automatically recorded by the System and stored in the user accounts and configuration databases.

Across all roles, whether Student, Counselor, or Admin, every flow converges into one unified ending: the System storing or updating data in the central database. This highlights the role of the Guidance Counseling Management System as the central handler of information, ensuring that all actions are securely processed and maintained. The process concludes once all interactions are completed and finalized by the system.

Entity-Relationship Diagram

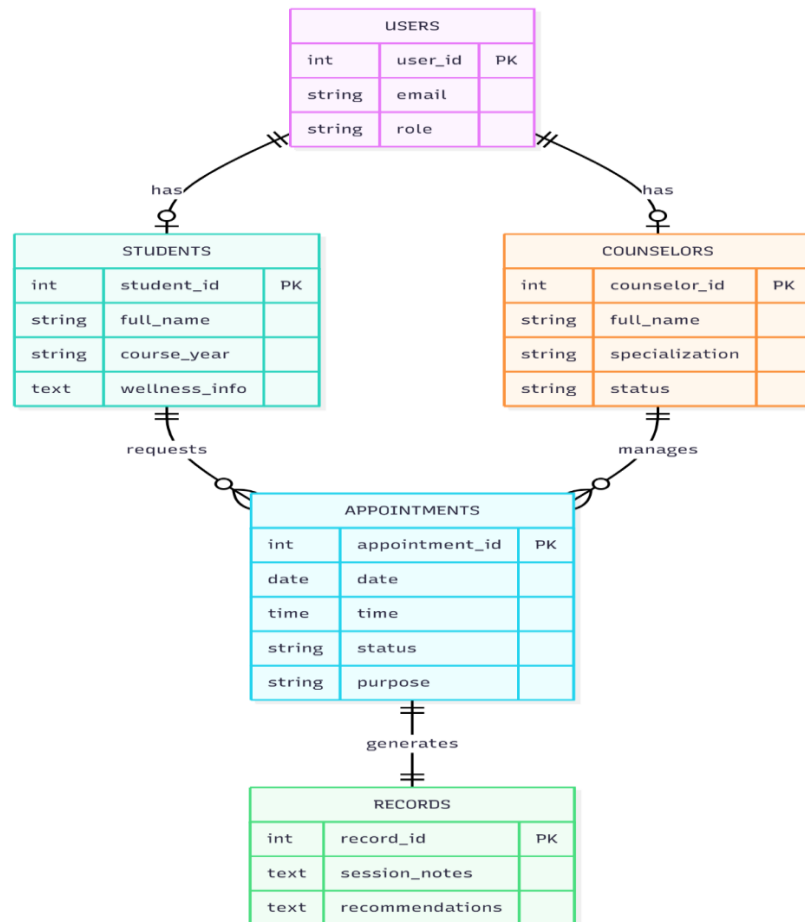


Figure 6. Entity-Relationship Diagram.

Figure 6 Entity-Relationship Diagram depicts the complete database schema of the Guidance Counseling Management System using Crow's Foot notation. The design adopts a supertype/subtype (inheritance) structure with the **USERS** entity acting as the parent table that stores common authentication data (user_id as primary key, email, and role). Through one-to-one identifying relationships, a user with role "student" is linked to exactly one record in the **STUDENTS** entity containing student-specific fields such as full_name,

course_year, and wellness_info, while a user with role “counselor” is linked to exactly one record in the **COUNSELORS** entity containing full_name, specialization, and status. This pattern ensures role-based access control, eliminates null values in role-specific columns, and maintains strict data normalization.

The **APPOINTMENTS** entity serves as the core transactional table that connects students and counselors. It establishes a many-to-one relationship with **STUDENTS** (one student can book multiple appointments) and another many-to-one relationship with **COUNSELORS** (one counselor can manage multiple appointments), with attributes including appointment_id (PK), date, time, status, and purpose to fully support the scheduling lifecycle.

To preserve confidentiality and enforce documentation compliance, each appointment produces exactly one corresponding entry in the **RECORDS** entity via a mandatory one-to-one relationship. The RECORDS table securely stores session_notes and recommendations, ensuring that sensitive counseling content remains directly associated with its originating appointment and cannot exist independently.

Overall, the ERD reflects a normalized, scalable, and secure database design that fully supports the system’s functional requirements: user authentication, appointment booking and assignment, session conduct, and confidential record-keeping — making it suitable for actual deployment in an academic institution’s guidance counseling office.

SYSTEM ARCHITECTURE

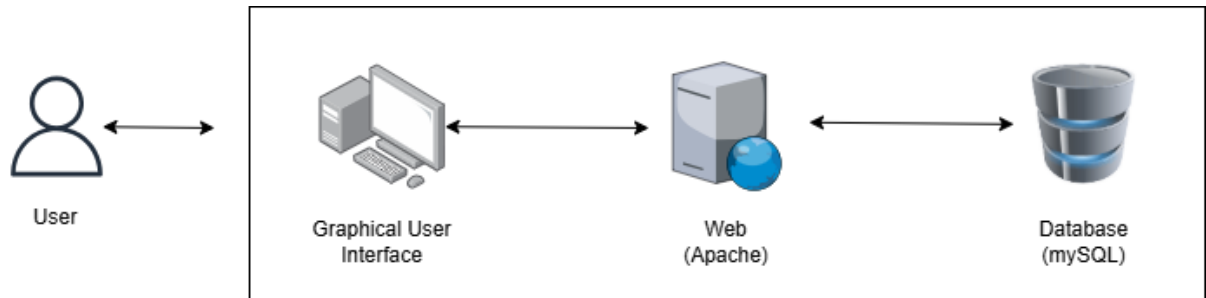


Figure 7. System Architecture.

The system architecture of the **Guidance Counseling Management System (GCMS)** follows a three-tier structure that ensures a seamless, secure, and efficient interaction between users and the application's core functionalities. At the forefront of this architecture are the users, who may take on different roles such as **students, counselors, or administrators**. These users interact with the system through a **Graphical User Interface (GUI)**, which serves as the visual and interactive layer of the application. Through this interface, users can perform a range of tasks such as registering an account, updating their profile, booking or rescheduling counseling appointments, and accessing assigned counselor sessions.

When a user initiates an action, the GUI captures the input and communicates it to the **web server**, which in this case is powered by **Apache**. The web server functions as the intermediary between the user interface and the database. It processes incoming requests,

executes the appropriate logic, and determines what kind of response or data is required. For example, if a student wants to view their scheduled counseling session, the web server interprets this request, applies the necessary business rules, and prepares a query for the database.

The final tier of the architecture is the **database**, built on **MySQL**, which acts as the central repository for all information in the system. This includes user account details, counseling schedules, student records, counselor session notes, notifications, and administrative data. The web server queries the database to either retrieve or update this information based on the user's interaction. Once the operation is completed—whether it is fetching scheduled appointments or saving a new counseling record—the database sends the result back to the server, which then communicates the outcome to the GUI for the user to see.

This three-tier architecture ensures that the GCMS operates with efficiency, scalability, and security, while maintaining a clear separation of concerns between the **presentation layer (GUI)**, **application logic (web server)**, and **data management (database)**.

SYSTEM PROTOTYPE

1. Authentication Module

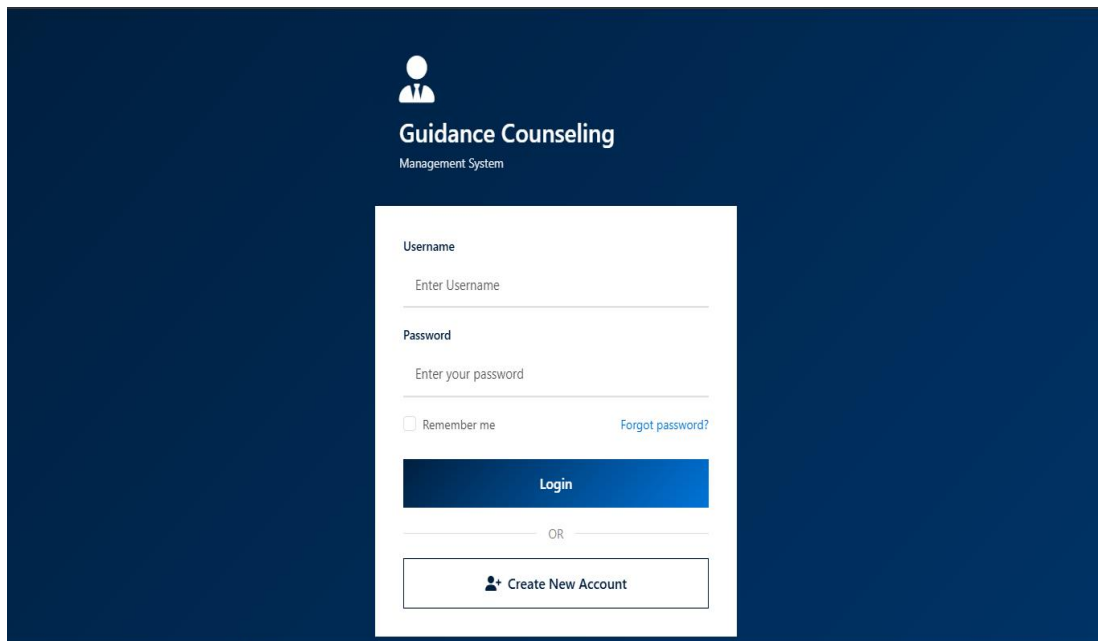


Figure 8. Login Page.

Figure 8 displays the Login Page, which serves as the secure entry point for all users. The interface requires users to enter their registered email address and password. The system uses a role-based redirection mechanism that automatically detects whether the user is a Student, Counselor, or Administrator upon authentication, directing them to their specific dashboard. It includes validation to ensure that only authorized personnel and registered students can access the system's sensitive data.



Student Registration

Create your account to access guidance counseling services



Personal Information

First Name *

Middle Name

Last Name *

mm/dd/yyyy

Gender *

Select Gender

Contact Number *

11 digit phone number (09123456789)



Academic Information

School ID *

Department *

Select Department

Course *

e.g., BS Computer Science

Year Level *

Select Year Level



Account Information

Create Username *

Password *

Confirm Password *


Submit Registration

Figure 9. Student Registration Page.

Figure 9 illustrates the Student Registration Page, designed for new users to create their accounts. The form collects essential demographic information, including the student's full name, identification number, course, year level, and institutional email. It enforces strong password requirements and validates that all mandatory fields are completed before

creating a new record in the database, ensuring that the system maintains accurate student profiles from the start.

2. Student Module

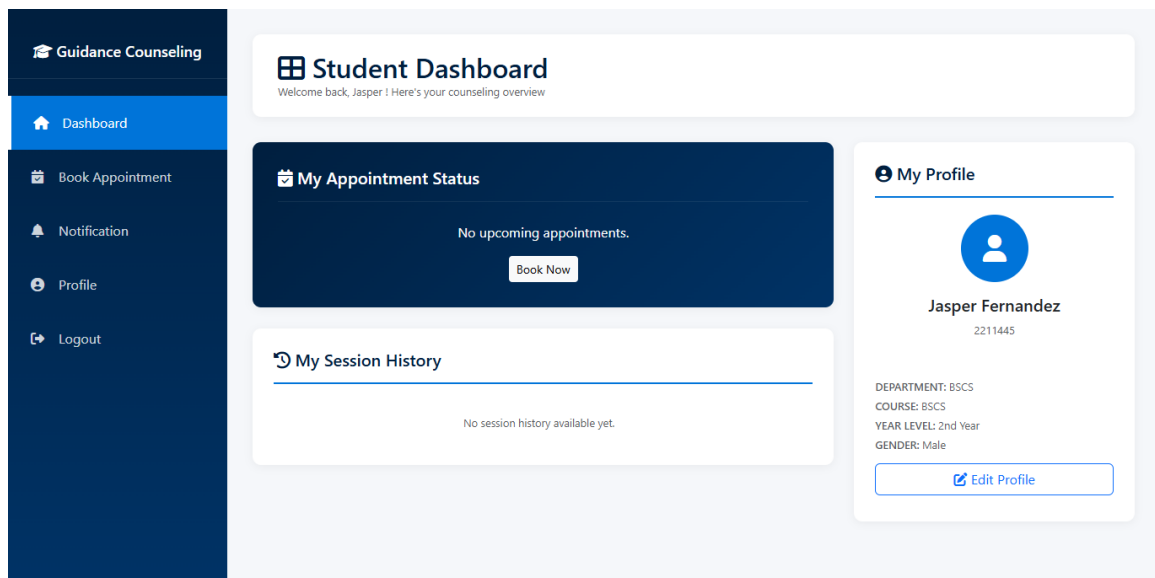
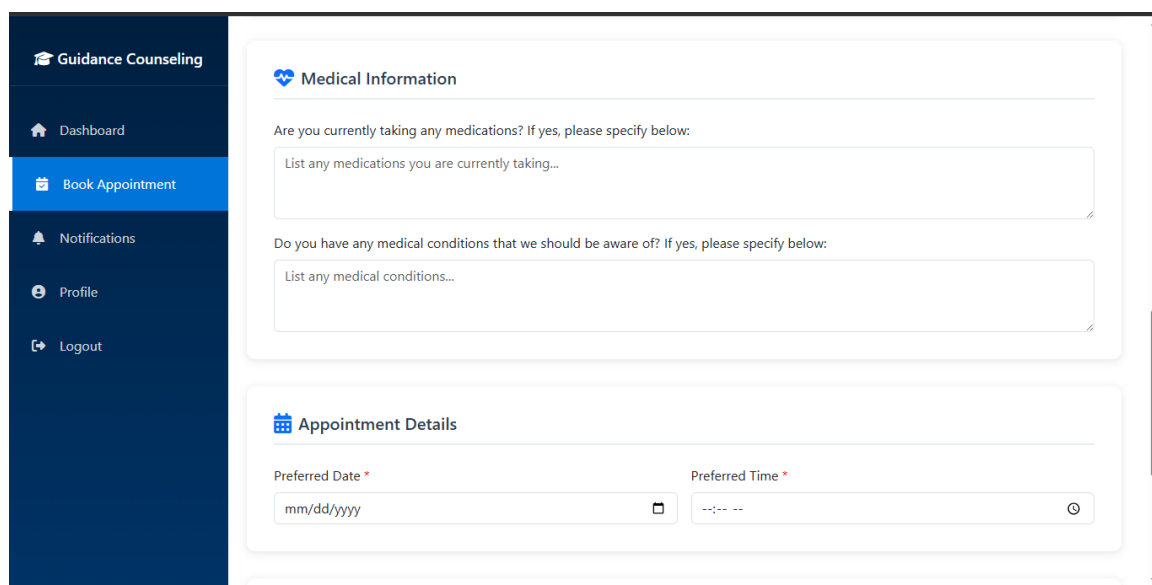


Figure 10. Student Dashboard.

Figure 10 shows the Student Dashboard, the central hub for student activities. It provides a summary of the student's current status, displaying counters for pending and confirmed appointments. The clean layout allows students to easily navigate to other modules using the sidebar menu.



Guidance Counseling

Dashboard

Book Appointment

Notifications

Profile

Logout

Medical Information

Are you currently taking any medications? If yes, please specify below:

List any medications you are currently taking...

Do you have any medical conditions that we should be aware of? If yes, please specify below:

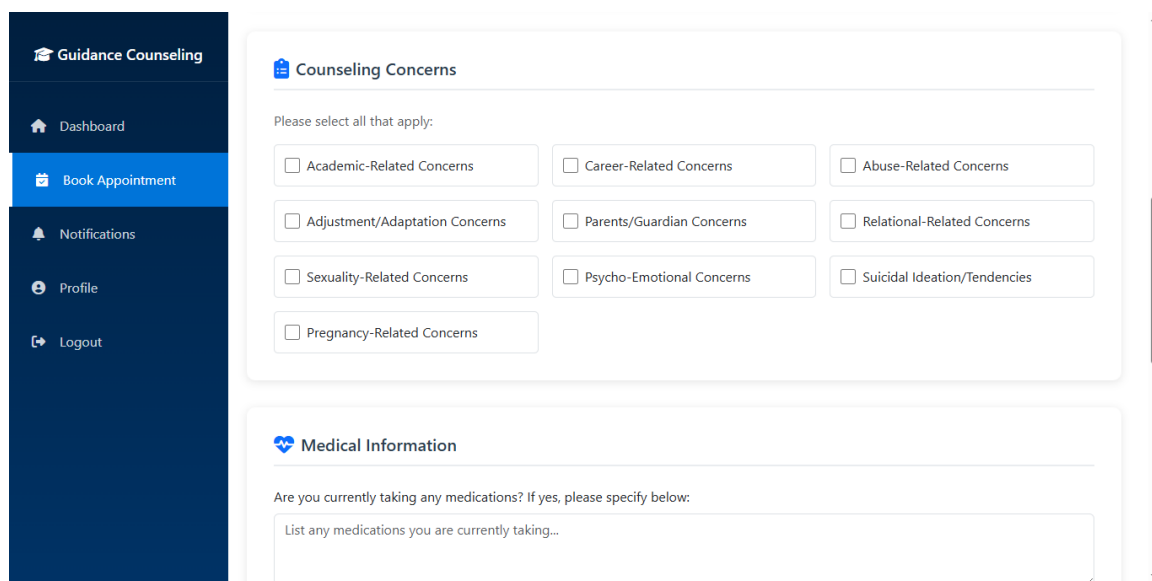
List any medical conditions...

Appointment Details

Preferred Date * Preferred Time *

mm/dd/yyyy --:-- --

Figure 11(a). Appointment Booking Module - Schedule Selection



Guidance Counseling

Dashboard

Book Appointment

Notifications

Profile

Logout

Counseling Concerns

Please select all that apply:

☐ Academic-Related Concerns ☐ Career-Related Concerns ☐ Abuse-Related Concerns

☐ Adjustment/Adaptation Concerns ☐ Parents/Guardian Concerns ☐ Relational-Related Concerns

☐ Sexuality-Related Concerns ☐ Psycho-Emotional Concerns ☐ Suicidal Ideation/Tendencies

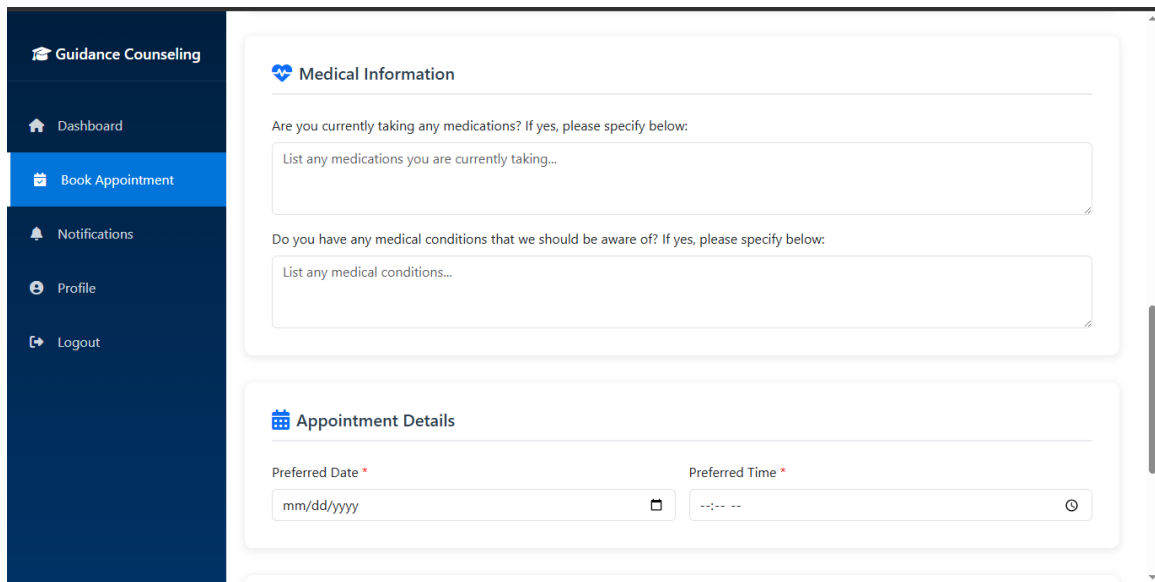
☐ Pregnancy-Related Concerns

Medical Information

Are you currently taking any medications? If yes, please specify below:

List any medications you are currently taking...

Figure 11(b). Appointment Booking Module - Wellness Intake Form (Physical)



Guidance Counseling

- Dashboard
- Book Appointment**
- Notifications
- Profile
- Logout

Medical Information

Are you currently taking any medications? If yes, please specify below:

List any medications you are currently taking...

Do you have any medical conditions that we should be aware of? If yes, please specify below:

List any medical conditions...

Appointment Details

Preferred Date * mm/dd/yyyy

Preferred Time * --:--

Figure 11(c). Appointment Booking – Wellness Intake Form(Intellectual)

Figure 11 presents the comprehensive Appointment Booking Interface, which guides the student through a multi-step request process. As shown in Figure 11a, the student begins by selecting a preferred date and time slot from the counselor's available schedule. Once a slot is chosen, the system requires the completion of a Wellness Questionnaire, as depicted in Figure 11b. This section gathers essential data regarding the student's Physical Wellness. The process continues in Figure 11c, where the student answers the remaining questions on Intellectual and Environmental Wellness before clicking the "Submit" button. This sequential design ensures that counselors receive a complete holistic profile along with the appointment request, allowing for better preparation before the session.

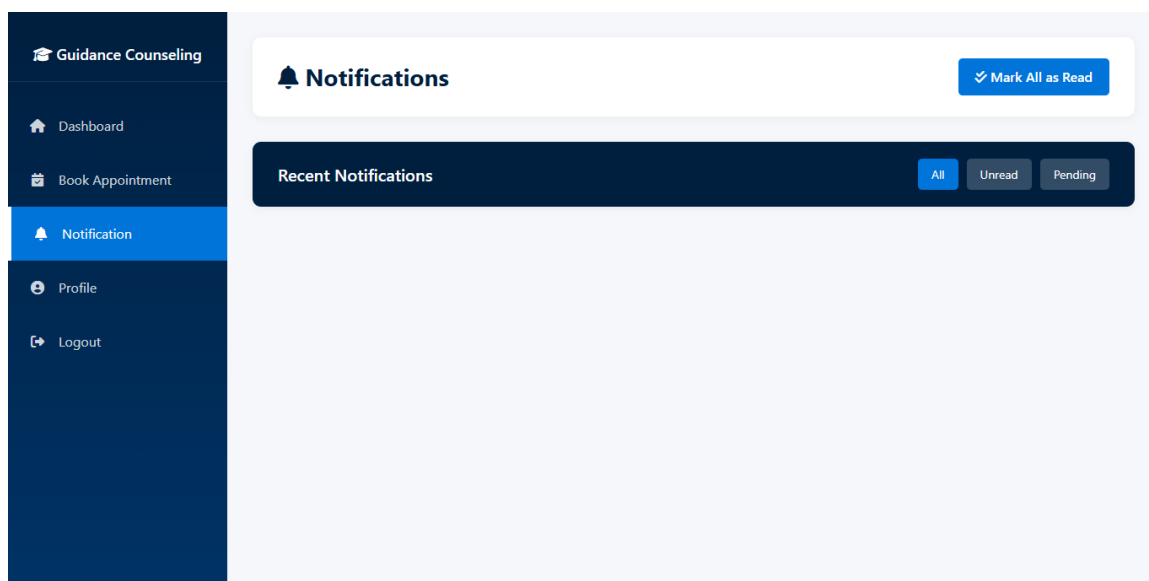


Figure 12. Student Notifications

Figure 13 illustrates the Student Notification Hub. This page aggregates all system alerts, such as appointment confirmations, cancellations, or rescheduling notices sent by the counselor. Each notification is time-stamped and marked as "read" or "unread," ensuring students never miss critical updates regarding their counseling sessions.

3. Counselor Module

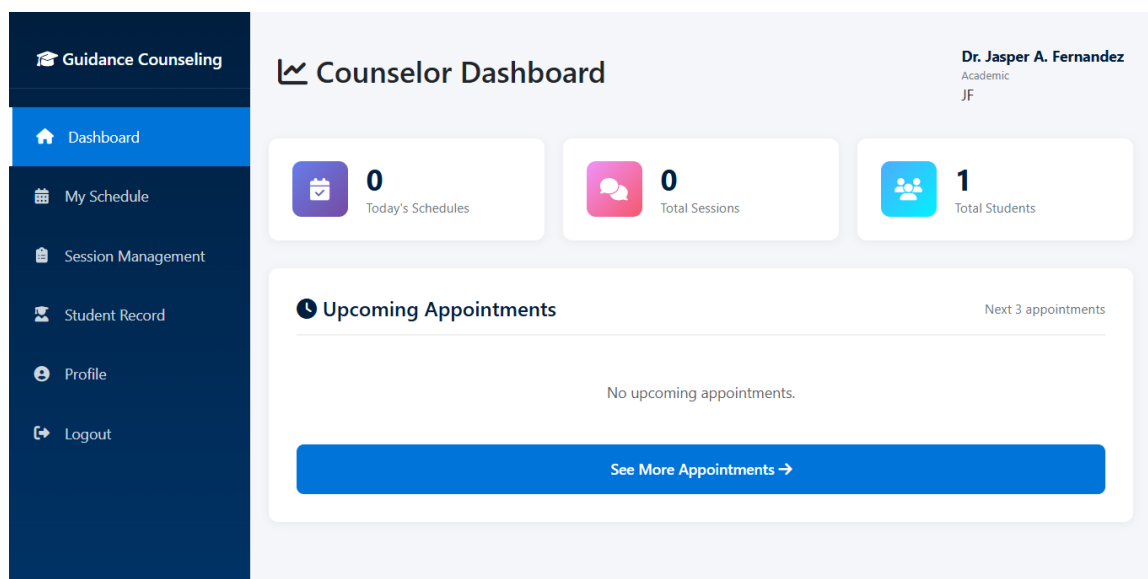


Figure 13. Counselor Dashboard

Figure 14 depicts the **Counselor Dashboard**, designed to help guidance staff manage their daily workload. It highlights key metrics such as the number of upcoming appointment requests and today's scheduled sessions, providing quick access to daily tasks.

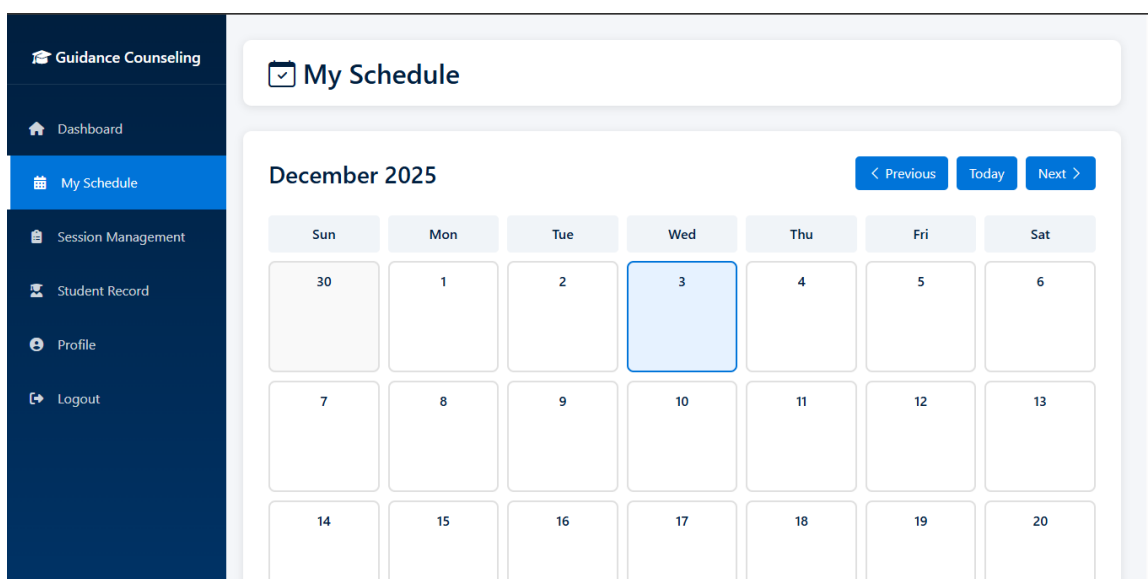


Figure 14. Counselor calendar schedule

Figure 14 serves as the primary time management tool for guidance counselors within the GCMS. As illustrated in the figure, the interface features an interactive, monthly calendar view (currently displaying December 2025) that allows counselors to visualize their workload at a glance. Key functionalities include navigation controls ("Previous", "Next", "Today") to toggle between months and a highlighted indicator for the current date to ensure immediate orientation. This module is integrated with the appointment database; dates with scheduled counseling sessions are visually distinct, allowing the counselor to click on specific days to view detailed appointment information, effectively preventing scheduling conflicts and ensuring organized service delivery.

Date & Time	Student	Purpose	Action
Nov 26, 2025 01:40 PM	Jasper Fernandez 2211445	Psycho-Emotional Concerns	▶ Start Session

Figure 15. Counselor Session Manager

Figure 15 Session Management page provides a structured interface for counselors to conduct active counseling sessions and review past cases. As shown in the figure, the module utilizes a tabbed interface to segregate data into two distinct views: "Active Sessions" and "Session History."

The Active Sessions tab displays a list of confirmed appointments sorted by date and time, providing key details such as the student's name, student number, and the purpose of the consultation. Each row features a "Start Session" button, which triggers a modal window for clinical documentation. This modal allows the counselor to record Attendance Status (Present/Absent), Session Notes, Recommendations, and Observations in real-time.

The Session History tab serves as an archival view, listing completed or cancelled sessions. It allows counselors to retrieve past records via a "View Notes" button, ensuring continuity of care by providing easy access to historical data and previous recommendations.

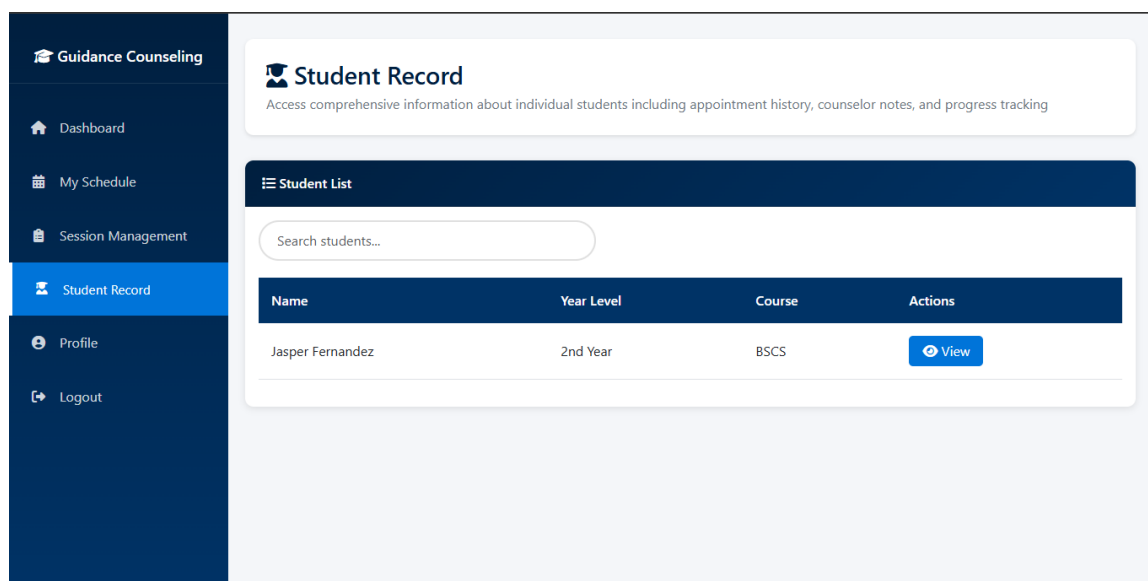


Figure 16. Counselor Student Record.

Figure 16 Student Record page acts as the central repository for student data within the counselor's portal. As depicted in the figure, the interface is divided into a searchable Student List and a detailed Student Profile Modal. The main view presents a responsive table listing all registered students, displaying key academic details such as their Name, Year Level, and Course. A search bar at the top allows counselors to quickly filter records by name or program.

Clicking the "View" button on any student row triggers a comprehensive modal that aggregates the student's entire counseling history. This detailed view provides immediate access to Personal Information, a chronological log of Counseling History (including dates, session types, and status), and the most recent Counselor Notes, Referrals (Recommendations), and Remarks (Observations). This design ensures that counselors have a holistic view of the student's progress and background before or during a session.

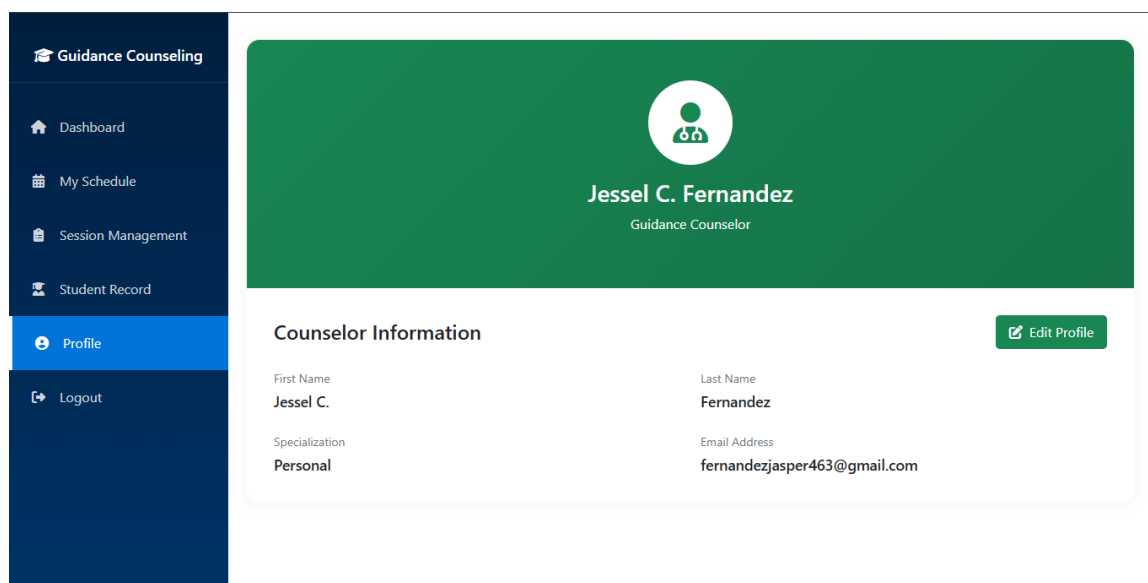


Figure 17. Counselor Profile.

Figure 17 Counselor Profile page provides a centralized interface for guidance counselors to view and manage their professional credentials and personal information within the system. As illustrated in the figure, the layout is divided into a Profile Header and a Details Section. The header features a visual avatar and the counselor's display name for immediate identification.

3. Administrator Module.

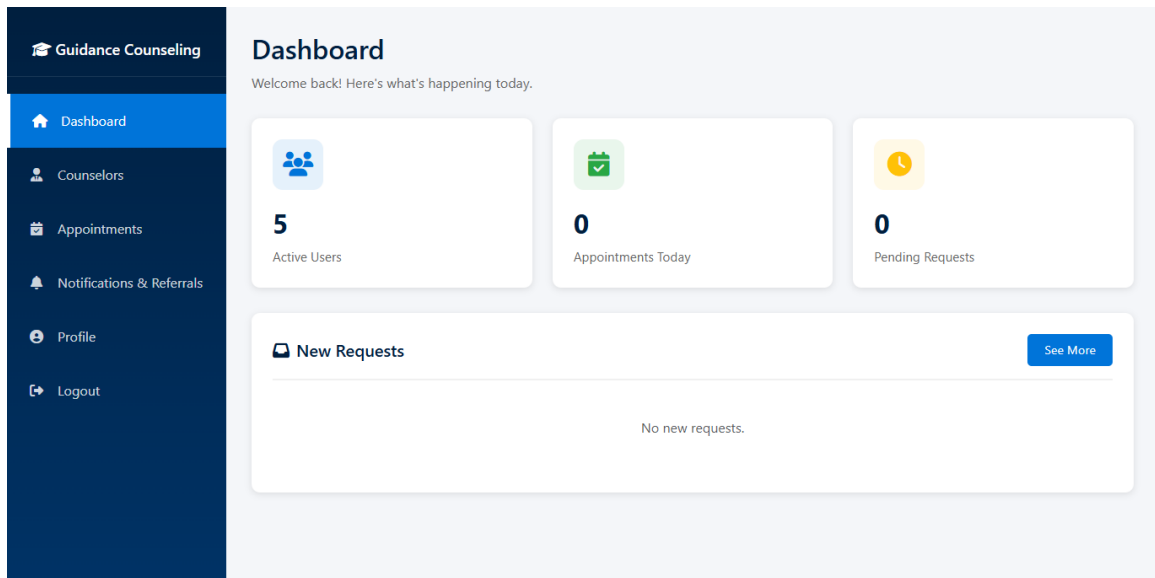


Figure 18. Admin dashboard.

Figure 18 Admin Dashboard serves as the central command center for the Guidance Counseling Management System, providing administrators with a high-level operational overview. As illustrated in the figure, the interface is designed with a Metrics Panel at the top, displaying real-time key performance indicators (KPIs) such as Total Active Users, Appointments Today, and Pending Requests. These counters provide immediate insight into the system's daily activity and workload.

Below the metrics, the "New Requests" section features a dynamic list of the most recent appointment applications. Each entry provides essential details—student name, date, time, and purpose—allowing administrators to quickly monitor incoming demand. The interface also includes quick-access navigation to core administrative modules like Counselor Management and Appointment Oversight, streamlining the workflow for system maintenance and user support.

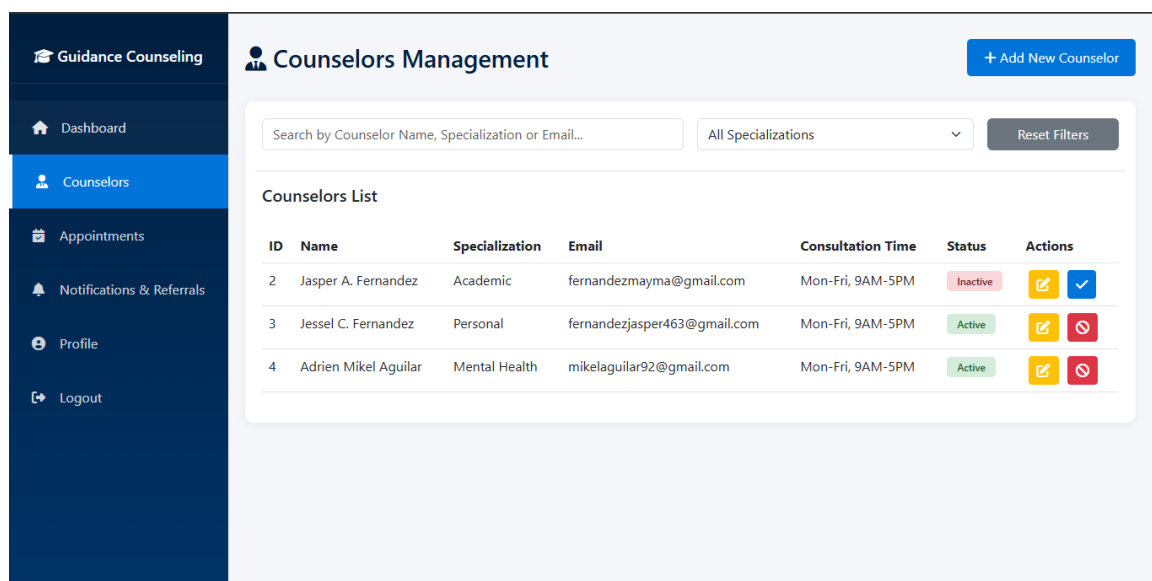


Figure 19. Admin Counselors Management Page.

The Admin Counselors Management Page serves as the centralized administrative hub for overseeing the guidance counseling workforce within the system. As illustrated in the figure, the interface presents a comprehensive Counselors List in a responsive tabular format, displaying critical professional details for each staff member, including their ID, Name, Specialization (e.g., Academic, Personal, Mental Health), Email, Consultation Time, and current Status. To facilitate efficient staff management, the page features a dynamic search bar and a "All Specializations" dropdown filter, enabling the real-time retrieval of specific counselor records. The system streamlines the onboarding process through the "+ Add New Counselor" button, which initiates the account creation workflow. Furthermore, the interface empowers administrators with direct control over account validity through the Actions column; this includes an Edit button (yellow pencil icon) for updating profile details and a Toggle Status button (blue check for activation or red ban

icon for deactivation). The current status of each counselor is visually indicated by color-coded badges—green for "Active" and red for "Inactive"—ensuring immediate clarity regarding staff availability.

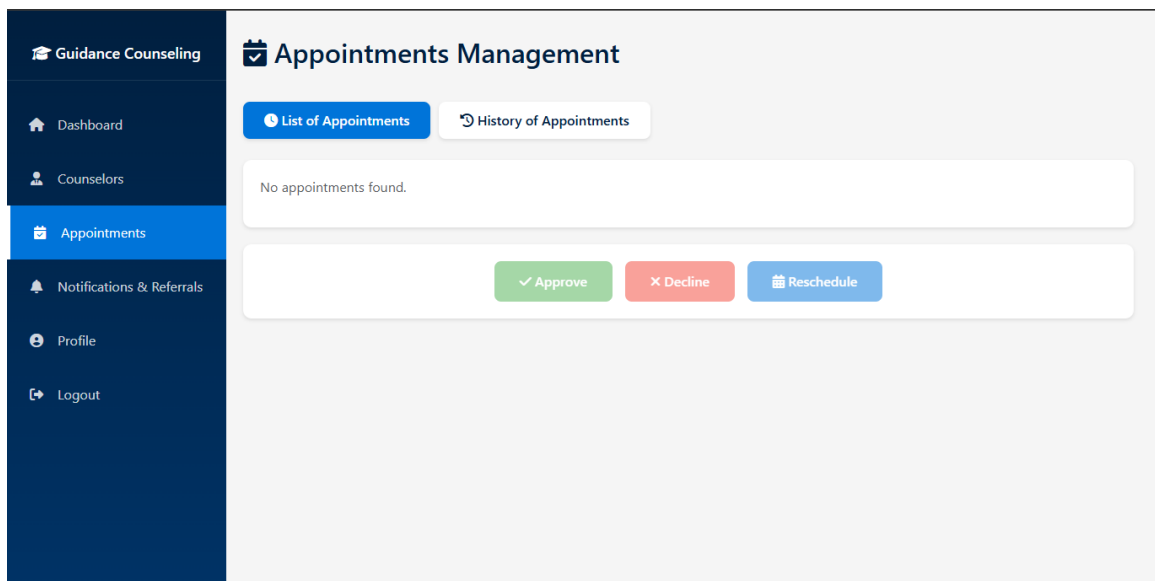


Figure 20. Admin Appointments Management Page.

The Admin Appointments Management page serves as the centralized control hub for overseeing the scheduling operations within the Guidance Counseling Management System. The interface is designed with a tabbed navigation system that segments data into two distinct views: "List of Appointments" for pending requests and "History of Appointments" for archival records of past sessions. The main view presents incoming student requests in a structured list, displaying critical details such as the student's name, requested date, time, and the purpose of the consultation. To facilitate efficient workflow, the page features a dynamic Action Bar containing "Approve," "Decline," and "Reschedule" buttons. These controls remain disabled until a specific appointment is

selected, preventing accidental modifications. Upon selection, administrators can process requests immediately; for instance, choosing "Reschedule" triggers a modal interface that allows for the selection of a new date and time, ensuring flexible conflict resolution. All actions are processed in real-time via AJAX, updating the appointment status instantly across the system without requiring a full page reload.

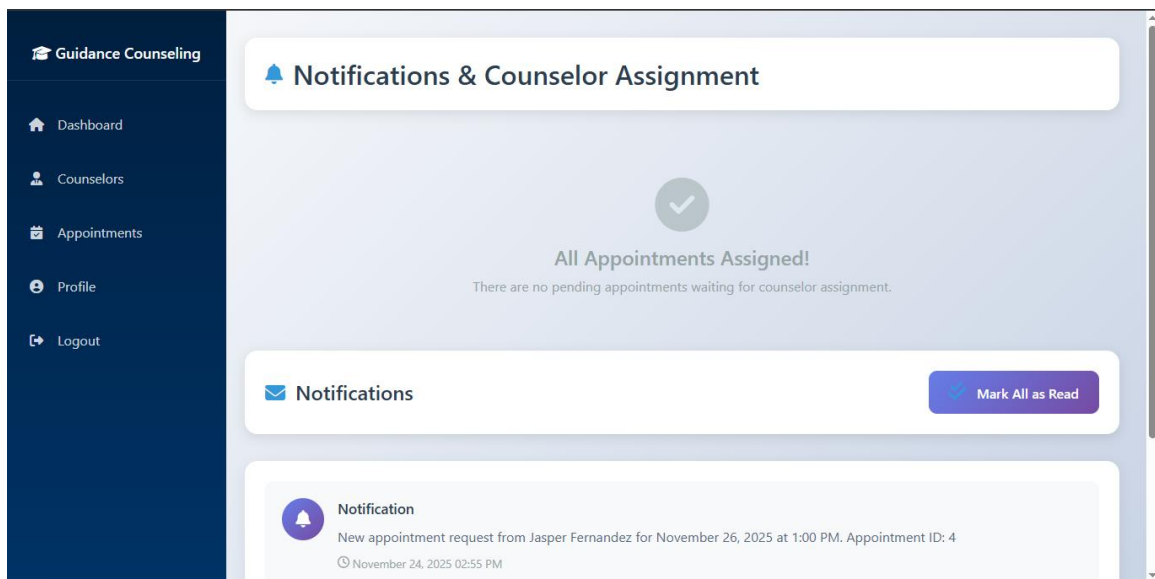


Figure 22. Admin Notifications & Counselor Assignment Page.

Figure 22 Notifications & Counselor Assignment page functions as a dual-purpose administrative command center designed to streamline workflow distribution and system awareness. As depicted in the prototype, the upper section features a "Pending Appointments" table that isolates appointment requests specifically requiring counselor assignment. This interface provides administrators with critical context—such as the student's name, course, requested schedule, and specific concerns—before they utilize the "Assign" action. Clicking this button launches a dedicated Assignment Modal, which

allows the administrator to search and select an active counselor from the staff database to handle the case. The lower section of the page acts as a real-time Notifications Feed, aggregating system alerts such as new booking requests or status updates. Each notification is time-stamped and visually distinguished by read/unread states, with a bulk action tool available to "Mark All as Read," ensuring that the administrator can efficiently track and clear the queue of system events.

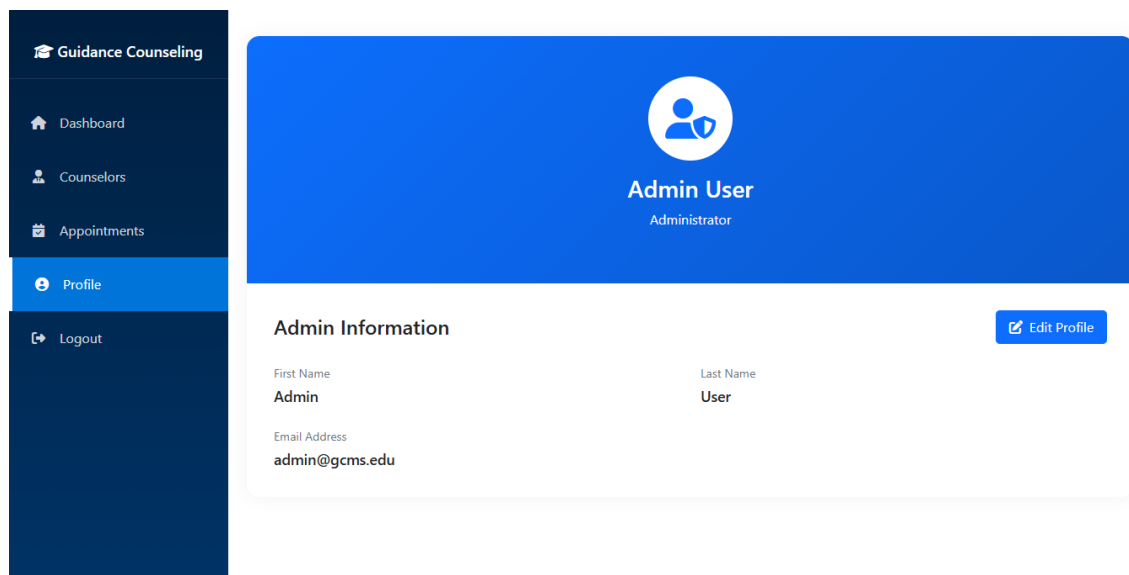


Figure 23. Admin Profile Management Page.

Figure 23 Admin Profile Management page provides a dedicated and secure interface for administrators to maintain their system credentials and personal information. As visually structured in the prototype, the page features a distinct Profile Header utilizing a professional gradient design and an avatar icon to clearly establish user identity and role authority. The interface employs a dual-mode interaction design to ensure data integrity: the default "View Mode" presents a read-only summary of the administrator's First Name,

Last Name, and Email Address, preventing accidental modifications. By clicking the "Edit Profile" button, the interface seamlessly transitions to "Edit Mode," converting static text into interactive input fields. This allows administrators to update their details efficiently, with changes being validated and committed to the database in real-time via asynchronous (AJAX) requests, eliminating the need for page reloads and ensuring a smooth user experience.

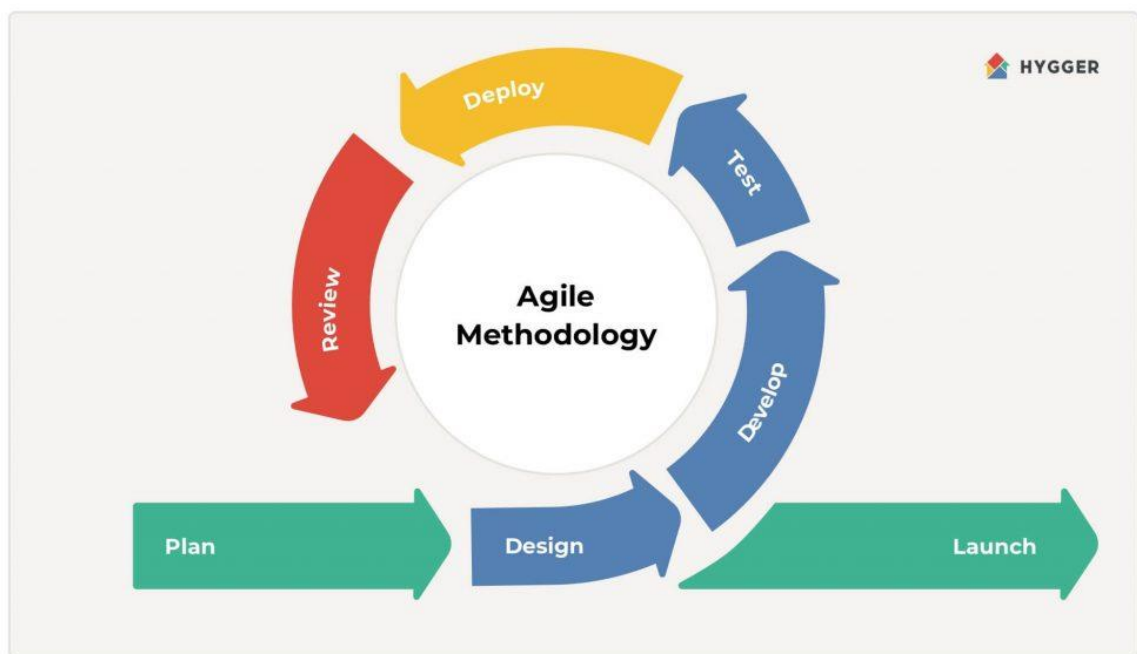
Development and Testing

Software Development Method

For the development of the **Guidance Counseling Management System (GCMS)**, the researchers adopted the **Agile Methodology** to ensure that the system was built in an iterative and user-centered manner. Unlike traditional linear models, Agile allowed flexibility in refining the system features based on the feedback of its intended users—students, counselors, and administrators of CatSU. Through incremental sprints, the essential modules such as appointment scheduling, counselor session management, and administrative account control were developed, tested, and improved until they met the identified requirements of the study.

In addition to development, this chapter also focuses on the **evaluation of the GCMS**. To assess the system's quality and acceptability, the researchers employed the **Descriptive Survey Method** guided by the **ISO/IEC 25010 software quality model**. This

framework measures software through eight key characteristics, including functionality, usability, reliability, and security. The evaluation results provided a clear indication of how well the system addressed the needs of its users and whether it could be adopted effectively within the university.



ImageSource: <https://hygger.io/guides/wp-content/uploads/2021/04/what-is-agile-1024x614.jpg>

Figure 21. Agile Methodology.

The development of the **Guidance Counseling Management System (GCMS)** was carried out using the **Agile Methodology**. Unlike traditional linear models, Agile emphasizes flexibility, collaboration, and incremental progress. This methodology was

chosen because it allowed the researchers to continuously refine the system based on user feedback from students, counselors, and administrators. Agile ensured that the GCMS evolved according to the real needs of its end-users while reducing risks associated with rigid, one-time development.

At the start of the development process, the researchers conducted requirement gathering and user story creation. Through interviews, surveys, and consultations, the team identified the challenges in the existing manual guidance counseling process. These requirements were translated into user stories such as: *“As a student, I want to schedule or reschedule an appointment so that I can meet my counselor at a convenient time”* and *“As a counselor, I want to record session notes so that I can maintain accurate student records.”* These user stories served as the foundation for prioritizing features in each Agile iteration or sprint.

The planning phase involved selecting the most essential features for the first sprint. High-priority functions such as user authentication, appointment scheduling, and counselor viewing of appointments were chosen because they formed the core of the system’s operations. This allowed the researchers to deliver a working prototype early in the development process, which could immediately be tested and evaluated by potential users.

During the development phase, the system was built incrementally. The team worked on one sprint at a time, developing modules such as Student Registration and Login, Appointment Scheduling, Counselor Session Management, and Admin User Account Management. After each sprint, the partially completed system was demonstrated

to users for feedback. This approach provided the flexibility to make changes based on suggestions, such as refining the user interface to be more intuitive or adjusting the flow of appointment notifications.

Testing was integrated into every sprint, ensuring that features were validated as soon as they were developed. Unit testing checked the correctness of individual modules, while integration testing ensured that modules worked together seamlessly. For example, the process of scheduling an appointment was tested end-to-end to confirm that the request made by a student was properly displayed on the counselor's dashboard and stored in the database. Bugs identified during testing were immediately corrected within the sprint before moving on to the next set of tasks.

The review and retrospective phase followed each sprint. In this phase, feedback was gathered from the stakeholders who interacted with the system. Students evaluated whether the scheduling feature was convenient, counselors checked the usability of session record management, and administrators tested the account management functions. The researchers discussed what went well, what challenges were encountered, and what improvements were needed in the next sprint.

Finally, the deployment phase was conducted after several completed sprints resulted in a functional and stable version of the GCMS. The system was deployed for trial use at CatSU, allowing real users to interact with it in an actual environment. Maintenance activities followed, focusing on minor refinements and enhancements based on additional

feedback. Since Agile is iterative by nature, the system remains open for continuous improvement and expansion even after deployment.

Through the use of Agile Methodology, the development of the GCMS was made more adaptable and user-centered. The iterative sprints ensured that the system's features were tested and validated continuously, reducing risks of failure. More importantly, Agile enabled active participation of students, counselors, and administrators throughout the development process, ensuring that the final product was aligned with their needs and expectations.

Evaluation

The evaluation of the Guidance Counseling Management System (GCMS) was conducted to measure its overall quality, usability, and acceptability among its intended users. To accomplish this, the researchers employed the Descriptive Survey Method, which made use of a structured questionnaire distributed to students, counselors, and administrators who were identified as the primary users of the system. The survey instrument was developed based on the ISO/IEC 25010 software quality model, which provides an internationally recognized framework for assessing software quality.

The evaluation focused on the eight characteristics outlined in ISO/IEC 25010. Functional suitability was assessed to determine whether the GCMS delivered the necessary features expected by its users, such as student appointment scheduling, counselor

record management, and administrative control of accounts. Performance efficiency measured the responsiveness of the system in processing login requests, scheduling transactions, and updates to the database. Compatibility referred to the ability of the system to operate seamlessly across different devices and web browsers commonly used in the university.

Usability was another crucial aspect, which examined the system's user interface, navigability, and ease of use for first-time users and frequent users alike. The reliability of the GCMS was assessed in terms of its ability to consistently perform its intended functions without system crashes or data errors, particularly during multiple simultaneous transactions. Security was evaluated to ensure that the system provided adequate protection for sensitive information, preventing unauthorized access to student and counselor records.

In addition, maintainability was measured to determine how easily the GCMS could be updated or modified when improvements or corrections were necessary. Finally, portability was considered in terms of the system's ability to adapt to different environments or be deployed in other academic institutions with minimal adjustments.

The results of the survey were tallied, analyzed, and interpreted to establish the system's overall acceptability. Based on user responses, the GCMS was found to be highly functional and reliable, with a user-friendly interface that met the needs of its intended users. Areas for improvement were also identified, particularly in enhancing system performance efficiency and strengthening security measures. These findings served as the basis for future refinements of the system.

Data Gathering Procedure

The study utilized both qualitative and quantitative approaches in gathering data to support the development and evaluation of the Guidance Counseling Management System (GCMS). During the development phase, data were collected through interviews, observations, and reviews of existing documents from the Guidance Office of CatSU. These activities provided the researchers with a clear understanding of the current manual process of managing appointments, counseling records, and student information. The insights gained from this stage allowed the researchers to identify the problems faced by students, counselors, and administrators, which were then translated into system requirements and user stories that guided the iterative development under the Agile methodology.

For the evaluation phase, the study employed a survey questionnaire anchored on the ISO/IEC 25010 software quality model. The instrument was validated by experts and administered to a sample group consisting of students, counselors, and administrators who tested the system during its trial deployment. The questionnaire aimed to measure the GCMS based on eight quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Respondents rated each item using a Likert scale, and the gathered data were tallied, analyzed, and interpreted to determine the overall quality and acceptability of the system.

Through this two-pronged data gathering approach, the study was able to capture both the requirements for development and the metrics for evaluation, ensuring that the GCMS was not only responsive to user needs but also aligned with international software quality standards.

Respondents of the Study

The respondents of this study were selected to ensure that the data gathered would be accurate and relevant to the objectives of the research. They include three groups within the school community: students, counselors, and administrative staff. Each group was chosen because of their direct involvement and varying perspectives that could contribute meaningful insights to the study. By including these respondents, the research aimed to capture diverse viewpoints and experiences.

In total, the population of possible respondents consisted of 27 individuals. From this number, a sample of 9 participants was drawn to represent each group proportionally. Specifically, 5 students, 3 counselors, and 1 administrative staff member were included in the study. This distribution ensures that the gathered data reflects a balanced representation of the target population. The summary of respondents is presented in the table below.

Table 7. Respondents of the Study

Types of Respondent	Total Population	Sample
Student	20	5
Counselors	5	3
Admin	2	1
	27	9

To effectively evaluate the objectives of the study, it was necessary to gather data from different groups within the school community. The respondents were composed of students, counselors, and administrative staff, with a total population of twenty-seven (27) individuals. From this population, a purposive sample of nine (9) participants was selected, consisting of five (5) students out of twenty (20), three (3) counselors out of five (5), and one (1) administrative staff member out of two (2).

This distribution was designed to ensure proportional representation of each group. By including individuals with distinct roles and responsibilities, the study was able to obtain diverse perspectives and insights that contributed to a more comprehensive understanding of the research problem. Such representation strengthens the reliability and validity of the findings, making them more reflective of the actual population being studied.

Statistical Tool

In analyzing the data collected from respondents, the study employed descriptive statistical tools to assess the system's usability, performance, and overall user satisfaction. These tools were selected for their effectiveness in summarizing feedback in a clear and quantifiable manner, thereby supporting evidence-based evaluation of the developed system.

Table 8. Likert Scale

Range	Quantitative Equivalent	Qualitative Equivalent
4.21 – 5.00	5	Strong Agree
3.41 – 4.20	4	Agree
2.61 – 3.40	3	Neutral
1.81 – 2.60	2	Disagree
1.00 – 1.80	1	Strongly Disagree

Arithmetic Mean (Average)

$\sum X$ = Sum of all observed values

N = Number of observations or respondents

One of the primary tools used was the weighted mean ($\bar{X} = \sum X / N$). This statistical method was applied to compute the average score for each evaluation criterion based on the respondents' ratings. It allowed the researchers to determine the overall sentiment toward specific aspects of the system, such as functionality, interface design, and communication features. The study also utilized frequency counts to determine how often particular responses or ratings appeared in the survey results. This helped identify trends and patterns in user feedback, particularly in areas where multiple respondents shared similar experiences or perspectives. To measure subjective responses, the study adopted a five-point Likert scale, which ranged from "Strongly Disagree" to "Strongly Agree." This scale was used to evaluate key 89 system attributes, including user satisfaction, ease of navigation, system reliability, and efficiency of the GIS and SMS functionalities. Together, these statistical tools provided a comprehensive analysis of user responses, enabling the researchers to draw meaningful conclusions about the system's effectiveness. The results helped identify both the strengths of the system and areas that may require further improvement prior to full-scale implementation.

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